

Assignment No. 10

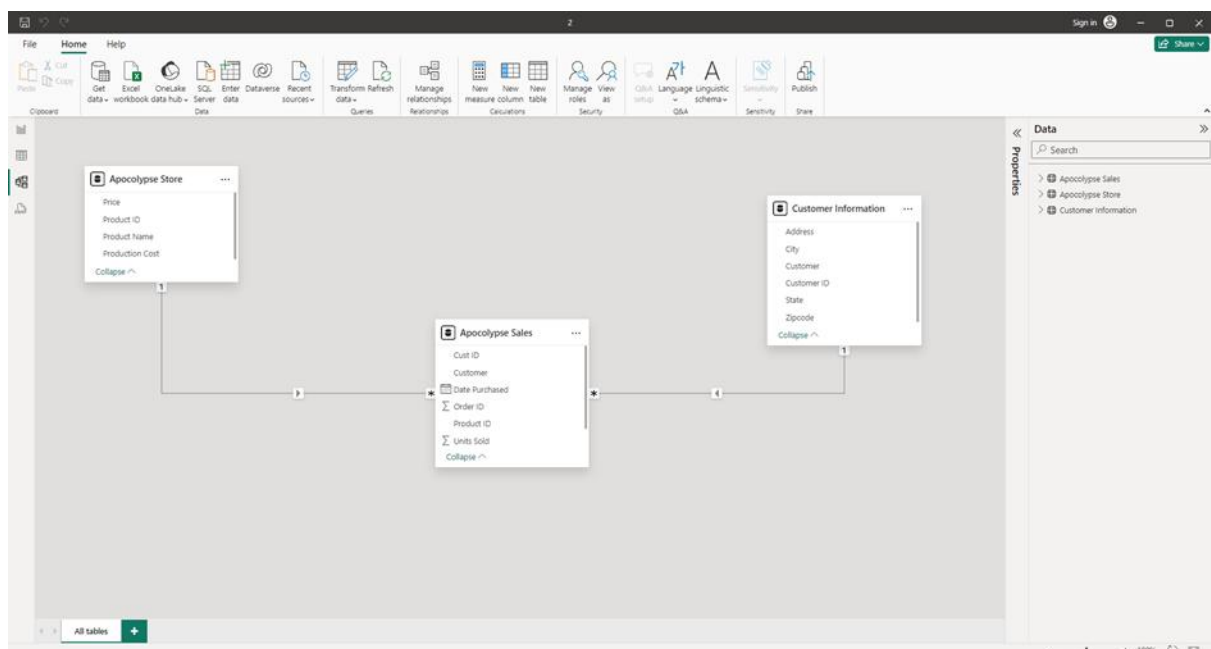
Experiment Title: Data Modeling

Problem Statement:

Download any dataset of CSV files, XML, JSON format and perform data modeling and create a basic dashboard using Tableau and PowerBI.

Theory:

1. Create relationship between the different tables in the given dataset.
2. Check for the unique attribute which is used for connection establishment.
3. Check the cardinality of the relationships and give reason for the same.
4. Delete the relationships between the Tables that were established by PowerBI by default and create new relationships between the tables considering the common attributes.



5. Create the DAX expressions for the following :
 - a. Count of sales
 - b. Sum of product sold

Customer	Sum of Units Sold
Uncle Joe's Prep Shop	951
Apocalypse Preppers United	828
Alex The Analyst Apocalypse Preppers	613
Prep4Anything Prepping Store	609
Total	3001

c. Profit made

```
1 profit = (SUM('Apocalypse Store'[Price]) - SUM('Apocalypse Store'[Production Cost])) * sum('Apocalypse Sales'[Units Sold])
```

Product Name	Price	Production Cost	profit
Weatherproof Jacket	79.99	30.59	13,091.00
Multitool Survival Knife	28.99	10.58	8,781.57
Nylon Rope	30.99	13.67	6,754.80
Stainless Steel Axe	45.50	32.45	4,567.50
Water Purifier	30.25	17.93	2,796.64
Backpack	39.99	26.92	2,770.84
Solar Battery Flashlight	26.49	13.41	2,380.56
Waterproof Matches	7.99	2.89	969.00
N95 Mask	2.75	1.01	596.82
Duct Tape	6.25	4.87	503.70
Total			4,34,754.87

d. DAX expression for SUMX for computing profit

```
1 profit_sumx = SUMX('Apocalypse Sales', ('Apocalypse Store'[Price] - 'Apocalypse Store'[Production Cost]) * 'Apocalypse Sales'[sum of unit sold])
```

Product ID	Product Name	Price	Production Cost	profit_col	profit_sumx
10001	Nylon Rope	30.99	13.67	434754.87	51977.32
10002	Waterproof Matches	7.99	2.89	434754.87	15305.1
10003	Stainless Steel Axe	45.5	32.45	434754.87	39163.05
10004	Backpack	39.99	26.92	434754.87	39223.07
10005	Multitool Survival Knife	28.99	10.58	434754.87	55248.41
10006	Solar Battery Flashlight	26.49	13.41	434754.87	39253.08
10007	Weatherproof Jacket	79.99	30.59	434754.87	148249.4
10008	Duct Tape	6.25	4.87	434754.87	4141.38
10009	Water Purifier	30.25	17.93	434754.87	36972.32
10010	N95 Mask	2.75	1.01	434754.87	5221.74

e. Create a new column for "Day of week"

1 day = WEEKDAY([Apocalypse Sales'[Date Purchased]],2)

Customer	Product ID	Order ID	Units Sold	Date Purchased	day	order_size
Uncle Joe's Prep Shop	10006	1904	32	01 January 2022	6	Big order
Alex The Analyst Apocalypse Preppers	10005	3966	68	02 January 2022	7	Big order
Apocalypse Preppers United	10004	7348	69	03 January 2022	1	Big order
Prep4Anything Prepping Store	10005	2409	48	04 January 2022	2	Big order
Uncle Joe's Prep Shop	10010	2045	70	05 January 2022	3	Big order
Alex The Analyst Apocalypse Preppers	10008	9151	6	06 January 2022	4	Small Order
Apocalypse Preppers United	10003	1065	59	07 January 2022	5	Big order
Prep4Anything Prepping Store	10003	3780	72	08 January 2022	6	Big order
Uncle Joe's Prep Shop	10001	1043	67	09 January 2022	7	Big order
Alex The Analyst Apocalypse Preppers	10007	7075	7	10 January 2022	1	Small Order
Apocalypse Preppers United	10001	6778	23	11 January 2022	2	Small Order
Prep4Anything Prepping Store	10008	3572	56	12 January 2022	3	Big order
Uncle Joe's Prep Shop	10006	4033	51	13 January 2022	4	Big order
Alex The Analyst Apocalypse Preppers	10005	2073	2	14 January 2022	5	Small Order
Apocalypse Preppers United	10010	7333	45	15 January 2022	6	Big order

f. Create a new column for "Order_Size", i.e if the Order_Size is greater than 25 then display "Big Order" else display "Small Order".

1 order_size = if([Apocalypse Sales'[sum of unit sold]]>25,"Big order", "Small Order")

Customer	Product ID	Order ID	Units Sold	Date Purchased	day	order_size
Uncle Joe's Prep Shop	10006	1904	32	01 January 2022	6	Big order
Alex The Analyst Apocalypse Preppers	10005	3966	68	02 January 2022	7	Big order
Apocalypse Preppers United	10004	7348	69	03 January 2022	1	Big order
Prep4Anything Prepping Store	10005	2409	48	04 January 2022	2	Big order
Uncle Joe's Prep Shop	10010	2045	70	05 January 2022	3	Big order
Alex The Analyst Apocalypse Preppers	10008	9151	6	06 January 2022	4	Small Order
Apocalypse Preppers United	10003	1065	59	07 January 2022	5	Big order
Prep4Anything Prepping Store	10003	3780	72	08 January 2022	6	Big order
Uncle Joe's Prep Shop	10001	1043	67	09 January 2022	7	Big order
Alex The Analyst Apocalypse Preppers	10007	7075	7	10 January 2022	1	Small Order
Apocalypse Preppers United	10001	6778	23	11 January 2022	2	Small Order

Conclusion: In conclusion, utilizing Tableau and Power BI for data modeling and dashboard creation enables efficient analysis and visualization of datasets in various formats. By preprocessing data and applying modeling techniques, users can uncover valuable insights. These insights are effectively communicated through interactive dashboards, facilitating informed decision-making and driving business success. Overall, these tools streamline data analysis processes, empower users to derive actionable insights, and enhance organizational efficiency.